

Intel PowerVia Technology: Backside Power Delivery for High Density and High-Performance Computing

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Abstract

This paper presents a high-yielding backside power delivery (BPD) technology, PowerVia, implemented on Intel 4 finFET process. PowerVia more directly integrates power delivery to the transistor as compared to published buried power rail schemes, enabling additional wiring resources on front side for signal routing. A fabricated E-core with >90% cell utilization showed >30% platform voltage droop improvement and 6% frequency benefit compared to a similar design without PowerVia. Transistor performance, reliability, and fault isolation capability is detailed.

Introduction

Aggressive transistor and interconnect scaling creates complex dependencies between standard-cell configurations and performance and area. Shorter cell heights enable lower power and smaller area designs at the expense of performance and increased routing complexity and congestion. Traditionally, power delivery and signal networks have been implemented with frontside wafer metallization. Buried Power Rail in the substrate acquires power and signals from frontside interconnects, consuming routing resources [Fig. 1a]. Transitioning the power delivery network (PDN) to the wafer backside eliminates power and signal network competition and provides improved area scaling, higher block utilization, and increased performance [Fig. 1b]. PowerVia extends BPD by more directly delivering power to the active devices [Fig. 1c] and boosts performance by enabling relaxed frontside metal scaling (e.g., M0 pitch relaxation). PowerVia enables a virtuous cycle of improvement: higher utilization enables shorter wires, requiring fewer buffers and in turn further area reduction for more compact and higher performing designs.

PowerVia Technology Features:

Fabrication uses Intel 4 as a foundation for transistor and frontside interconnect processing. Due to the signal routing improvements afforded from PowerVia and the BPD, a modified 210nm cell height library is utilized, including a relaxed M0 pitch for improved standard-cell performance. Figure 2 details key technology dimensions for Intel 4 with and without PowerVia. A TEM [Fig. 3] highlights the technology stack with frontside metals, transistors, and backside metals. The process flow begins with standard fabrication steps to build the finFET transistors [1]. PowerVias are integrated to be coplanar with the source/drain contacts [Fig. 4a] and a 14-layer frontside interconnect stack is built with dimensions optimized for signal routing. The wafer is bonded to a carrier wafer and thinned to reveal the PowerVia connection on the backside of the wafer [Fig. 4b-c]. Bond-induced wafer distortion is controlled to enable cell-scaled backside interconnects at high yield. Finally, low resistance metal layers for power delivery are fabricated to form the backside PDN [Fig. 4d].

Device and Reliability:

FinFET devices with PowerVia processing show no impact on FET behavior, with Id-Vg and Id-Vd characteristics matching the reference Intel 4 process on both N/P MOS devices [Fig. 5]. Impacts of PowerVia on FEOL reliability has been widely investigated for relevant reliability degradation modes, such as TDD, BTI, and HCI. PowerVia architecture delivers design benefits with no added reliability penalty, as exemplified by the NBTI comparison on similar transistors with PowerVia versus the reference flow [Fig. 6]. Through process and metallurgical optimizations, excellent electromigration reliability has been realized in the crucial metallization linking the BPD network to the frontside signal network [Fig. 7]. Backside metallization presents thermomechanical reliability as a critical aspect of PowerVia development, with PowerVia meeting all standard JEDEC thermomechanical stress requirements with zero failures.

IR drop and performance:

IR drop is a bottleneck in highly scaled interconnect stacks, particularly for high-performance and low supply voltage designs, forcing designers to reserve coveted frontside routing resources to construct larger, more robust power grids [2]. Reduced metallization in the power pathways from the backside PDN significantly lowers the IR drop by 5x from the wafer bump to transistor (simulated), and >30% from package to transistor (on-chip monitors). More than 6% frequency is achieved on an E-core using PowerVia as compared to the Intel 4 reference design [Fig. 8]. High cell utilization >90% is demonstrated on an E-core with 2.9mm² area [Fig. 9]. A 10x8mm² window with 95% cell utilization shows densely packed standard cells [Fig. 10].

Yield and Failure Analysis:

Intel 4 with integrated PowerVia tracks the defect density of Intel 4 with minimal offset in process maturity [Fig. 11]. No defect issues specific to PowerVia have been identified to prevent yield from achieving HVM requirements. Failure debug and defect isolation is complex due to the presence of metallization on both sides of the wafer. Modified conventional and novel debug techniques have been developed to enable successful localization of hard defects and speed path issues [Fig. 12].

Summary

PowerVia on Intel 4 efficiently routes a backside PDN to the transistor with no adverse impacts to transistor performance, reliability, or yield. Benefits of lower supply droop, improved area scaling, and higher design frequencies are realized on an E-core versus an Intel 4 reference.

References

- [1] B. Sell, VLSI, 2022, pp.249.
- [2] D. Prasad, IEDM Tech. Dig., 2019, pp. 446-449

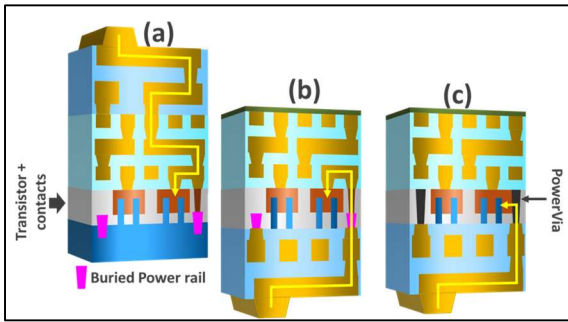


Fig. 1. Comparison b/w (a) Buried Power Rail without BS-PDN. (b) Buried Power Rail with BS-PDN. (c) PowerVia with BS-PDN.

	Intel 4	Intel 4 + PowerVia
Contacted Poly Pitch (nm)	50	50
Fin pitch (nm)	30	30
M0 pitch (nm)	30	36
#front-side layers	15+RDL	14
#back-side layers	n/a	4+RDL
HP library height (nm)	240	210

Fig. 2. Comparison of main technology features for Intel 4 and Intel 4+PowerVia.

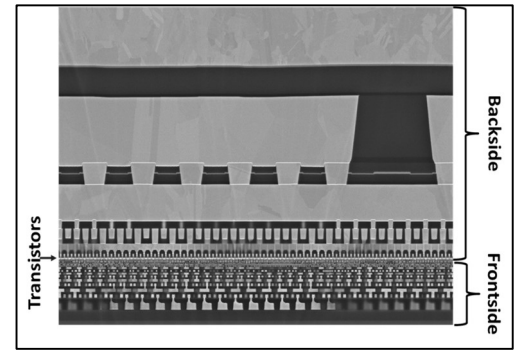
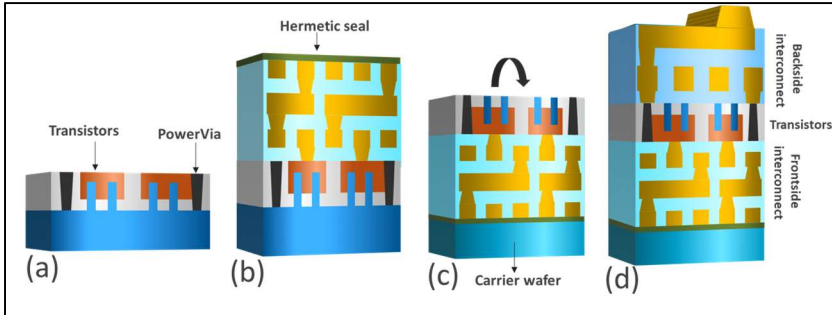


Fig. 3. Intel4+PowerVia TEM showing backside PDN, active transistor, and frontside signal metals.

Fig. 4. Schematic of process flow details. (a)



finFET patterning + PowerVia integration with S/D contacts. (b) BE interconnects optimized for signal routing. (c) Carrier bond → wafer thinning → PowerVia reveal. (d) BE interconnects optimized for power delivery.

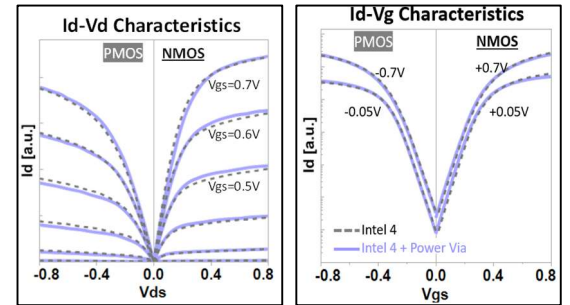


Fig. 5. Transistor Id-Vg and Id-Vd characteristics for both N/P MOS are matched between Intel 4 and Intel 4+PowerVia.

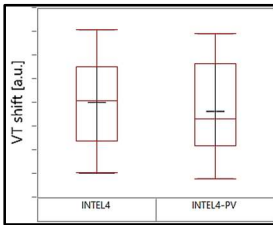


Fig. 6. PFET NBTI Vt shifts for Intel 4 without (left) and with (right) PowerVia.

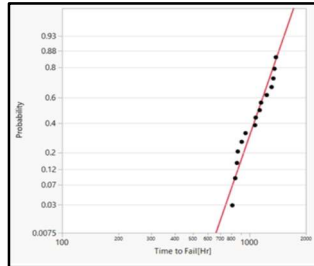


Fig. 7. Electromigration between PowerVia to backside PDN transition meeting 10-yr operation requirements.

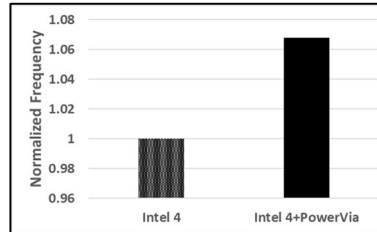


Fig. 8. Power delivery stress patterns show 6% faster core frequency using PowerVia versus reference frontside PDN of Intel 4.

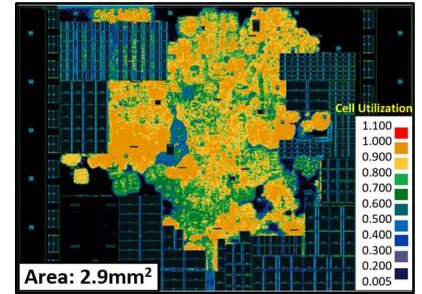


Fig. 9. E-core implemented in PowerVia technology enables high cell utilization (>90%).

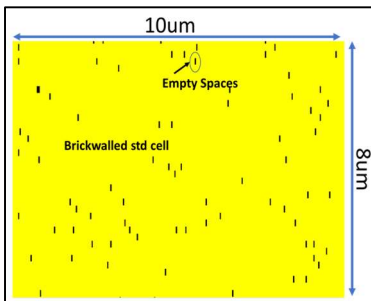


Fig. 10. Zoomed in view of a region with ~95% cell utilization.

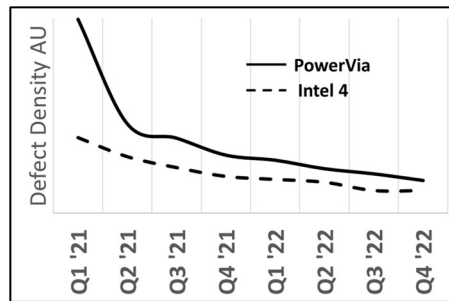


Fig. 11. PowerVia defect density tracks Intel 4 maturity with ~2 quarter offset.

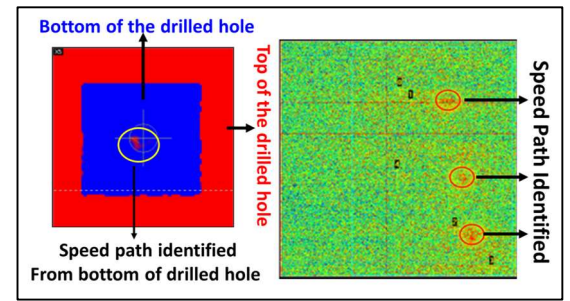


Fig. 12. Fault isolation localizing speed path failures using two different techniques.